🔋 REFERENCE STORAGE & WIRING MANUAL

New Energy Grid

Solis Retrofit / Hybrid Inverters & Easyway 5200 (UNIV5200) Battery Guide

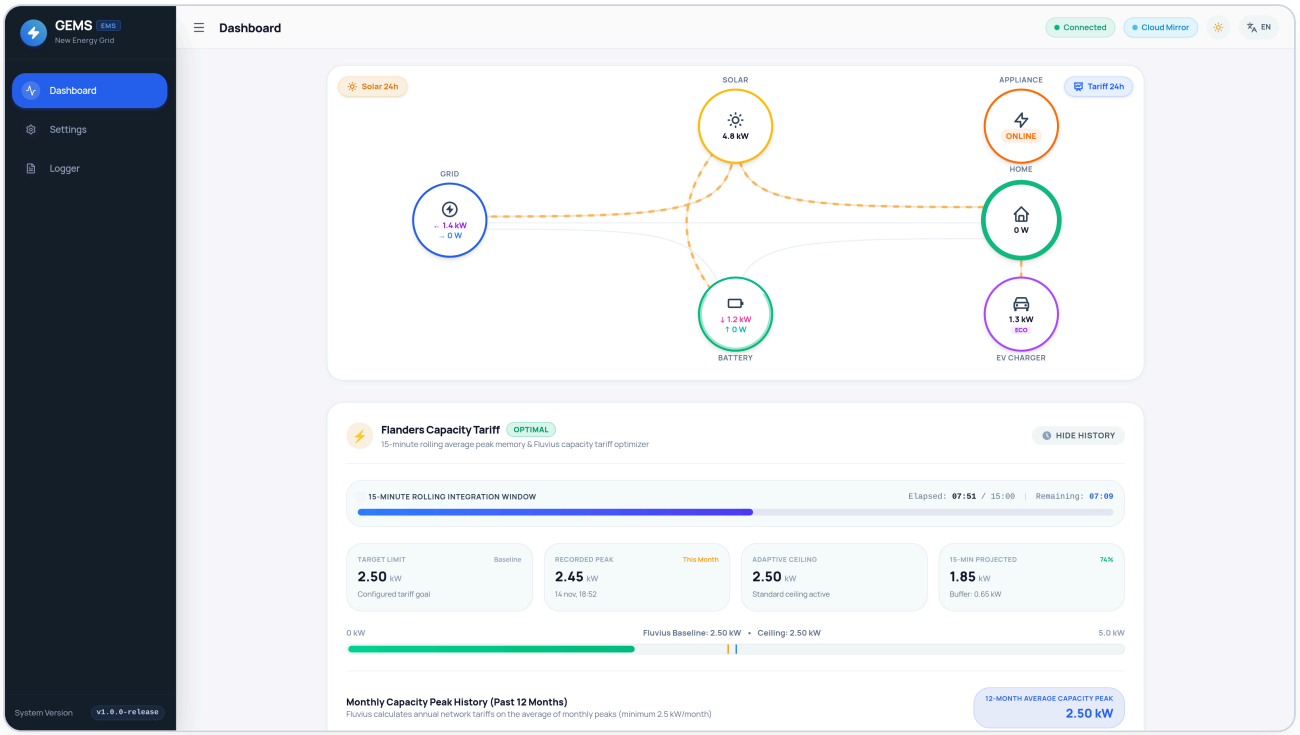
Complete electrician & homeowner manual covering **physical DC power, CAN BMS & parallel DIP-switch wiring** between the **Easyway UNIV5200 (51.2V LiFePO$_4$)** battery and **Solis Retrofit / Hybrid** inverters, **Eastron SDM72D-M & SDM630-M** smart meter wiring & setup, exact **SolisCloud App** settings, and optimal **GRID-EMS (GEMS)** configuration.

DOCUMENT VERSION
v9.36.7-release (Official Tag)

GEMS PRIMARY REFERENCE TEMPLATES
solis_hybrid • solis_retrofit_battery • eastron_sdm630 • eastron_sdm72_x96

SUPPORTED SOLIS LV (48V) INVERTERS
RAI-3K-48ES-5G • S6-EA1P-L (Retrofit) • S5-EH1P-L • S6-EH1P-L • S6-EH3P-L

BATTERY & SMART METER HARDWARE
Easyway UNIV5200 (51.2V 5.22 kWh, 1–16×) • Eastron SDM72D-M & SDM630-M



> ⭐ **Primary Reference Storage Platform in GRID-EMS**
>
> **Solis** hybrid and AC-coupled retrofit inverters paired with 51.2V modular **Easyway UNIV5200** LiFePO$_4$ packs serve as the primary reference energy storage architecture in GEMS. Whether upgrading an existing solar home via an AC-coupled Solis Retrofit inverter or commissioning a brand-new 1-phase/3-phase Solis S6 Hybrid plant, following this guide guarantees safe balanced DC currents, zero BMS CAN communication faults, flash-safe volatile Modbus dispatch, accurate Flanders 15-minute peak shaving, and intelligent Day-Ahead spot arbitrage.

© New Energy Grid & The Charge Grid — Proprietary Field Documentation v9.36.7 Target Platform: GRID-EMS Edge Controller (Raspberry Pi ARM64)

Section 1: Hardware Compatibility, Sizing & Voltage Guardrails

The **Easyway 5200 Energy Storage Battery (`UNIV5200`)** is a **51.2 V nominal (48V class) Low-Voltage Lithium Iron Phosphate ($LiFePO_4$)** module rated at `102 Ah` (`5.22 kWh` total energy / `~4.70 kWh` usable at 90% DoD). Before mounting or wiring hardware, verify strict voltage compatibility and pack-to-inverter power sizing.

> 🛑 **CRITICAL VOLTAGE GUARDRAIL: Low-Voltage (-L / 48V) Solis Models ONLY!**
>
> - **NEVER connect Easyway UNIV5200 modules in series!** All multi-battery expansions (2 to 16 modules) must be wired **strictly in 51.2 V DC parallel** (`P+` to `P+`, `P-` to `P-`).
> - **NEVER connect UNIV5200 batteries to High-Voltage (`-H`) Solis inverters** (such as `S6-EH3P(5-20)K-H` or `RHI-3P-HVES`), which operate at `160–600 Vdc`. Always verify the `-L` or `48ES / 48V` suffix on the Solis nameplate!

1.1 Easyway UNIV5200 Technical Specification Summary

Parameter	UNIV5200 Specification	Parameter	UNIV5200 Specification
Cell Chemistry	16S Prismatic $LiFePO_4$	**Recommended Charge / Discharge**	`50 A` continuous (~`2.56 kW` / pack)
Nominal Voltage & Capacity	`51.2 Vdc` • `102 Ah` (`5.22 kWh`)	**Max Continuous / Peak Current**	`100 A` max (~`5.12 kW`) / Peak 10s
Operating Voltage Window	`44.8 Vdc – 57.6 Vdc`	**Parallel Scalability**	Up to **16 packs** (`83.52 kWh`)
Recommended Depth of Discharge	`90 % DoD` (`10%` floor SOC)	**Ingress, Cooling & Temp Range**	`IP20` indoor, passive; Charge `0–55 °C`

1.2 Compatible Solis LV Inverters & Recommended UNIV5200 Pack Sizing

Because each `UNIV5200` BMS recommends `50 A` continuous current (~`2.56 kW` per module), pairing a single 5.22 kWh pack with a 5 kW or 10 kW inverter causes the BMS to dynamically cap continuous charge/discharge around ~2.56 kW. Use the table below to match inverter power, minimum recommended packs, DC cable gauge, and external DC breaker rating:

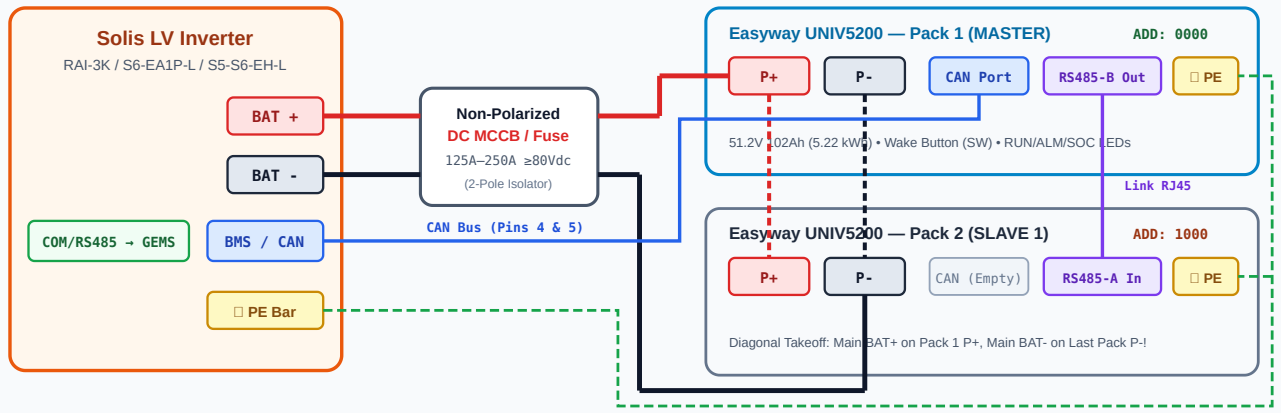
Solis LV Inverter Series	Type & GEMS Template	Max Battery Current	Recommended UNIV5200 Stack	Main DC Cable & Breaker
Solis RAI-3K-48ES-5G *(3.0 kW, 1-Phase AC)*	AC Retrofit (No PV) `solis_retrofit`	`60 A` (~3.0 kW)	**1 – 3× UNIV5200** *(5.22 – 15.66 kWh)*	`25 mm²` copper **100 A** DC MCCB
Solis S6-EA1P(3.6–6)K-L *(3.6 / 5.0 / 6.0 kW, 1-Phase AC)*	AC Retrofit (No PV) `solis_retrofit`	`75 A – 125 A`	**2 – 4× UNIV5200** *(10.44 – 20.88 kWh for full 5kW+)*	`35 mm²` copper **125 A – 160 A** DC MCCB
Solis S5-EH1P(3–6)K-L *(3.0 – 6.0 kW, 1-Phase Hybrid)*	PV + Storage Hybrid `solis_hybrid`	`62.5 A – 100 A`	**2 – 4× UNIV5200** *(1× works up to ~2.56 kW cont.)*	`25 – 35 mm²` copper **125 A** DC MCCB
Solis S6-EH1P(3–8)K-L-PLUS/PRO *(3.0 – 8.0 kW, 1-Phase Hybrid)*	PV + Storage Hybrid `solis_hybrid`	`70 A – 190 A`	**2 – 6× UNIV5200** *(≥3 packs for 8 kW continuous)*	`35 – 50 mm²` copper **160 A – 250 A** DC MCCB
Solis S6-EH3P(5–10)K-L & **NV-YD-L** *(5.0 – 15.0 kW, 3-Phase LV Hybrid)*	3-Phase LV Hybrid `solis_hybrid`	`120 A – 290 A`	**3 – 8× UNIV5200** *(≥3 packs via DC Busbar)*	`50 – 70 mm²` (or dual 35 mm²) **200 A – 300 A** DC MCCB

Section 2: Step-by-Step Wiring Manual — Easyway UNIV5200 to Solis Inverter

All wiring must be performed with the Solis AC breaker, PV DC isolator, external DC battery breaker, and all Easyway `UNIV5200` pack switches strictly **OFF**. Because 51.2 V battery banks carry high DC currents (up to 100–250 A), meticulous lug crimping, equalized parallel cable paths, and exact RJ45 CAN pin matching are essential.

2.1 DC Power, Protective Earth & Parallel Balance Wiring

Master Wiring Schematic: 2× Parallel Easyway UNIV5200 (Diagonal Balance) + DC MCCB + CAN BMS to Solis Inverter



1. **Single UNIV5200 Module (5.22 kWh):** Run equal-length `25 mm²` (or `35 mm²`) fine-stranded flexible copper cables from `P+` (Red) and `P-` (Black) through an external **non-polarized 2-pole DC MCCB breaker** (`100 A – 125 A`, rated `≥ 80 Vdc`; never use polarized AC breakers!) directly to the Solis inverter `BAT+` and `BAT-` terminals.
2. **2 to 3 Parallel UNIV5200 Modules (Diagonal / Cross-Takeoff Rule):** Link `P+` to `P+` and `P-` to `P-` between modules using identical-length bridge cables. Connect the main positive cable going to the Solis inverter onto **Pack 1 (Master)** `P+`, and connect the main negative cable going to the Solis inverter onto the **LAST Pack's** `P-` (Pack 2 or Pack 3). This equalizes total loop resistance across every pack so Pack 1 does not age prematurely.
3. **4 to 16 Parallel UNIV5200 Modules (Copper Busbar Rule — Mandatory > 3 Packs):** Do *not* daisy-chain high currents across 4+ battery terminals. Instead, install rated tin-plated **Positive and Negative Copper DC Busbars** (`250 A – 400 A`). Run identical-length `25 mm²` fused branch cables from each individual `UNIV5200` to the busbars, then run main `35 mm² – 70 mm²` cables from the busbars through the main DC MCCB to the Solis inverter.
4. **Terminal Torque & Protective Earth (`PE`):** Match your UNIV5200 hardware revision terminals (firmly click **Amphenol SurLok Plus** plugs until the lock tab latches, or torque `M6 / M8` hydraulic hex lugs to `5–8 Nm` with spring washers—never trap insulation under a lug!). Bond every `UNIV5200` chassis ground screw (`⏚`) to the Solis PE bar using `≥ 6 mm²` green/yellow earth wire.

2.2 BMS CAN Cable Pinout (Master UNIV5200 to Solis BMS Port)

Closed-loop communication between the Master `UNIV5200` and the Solis inverter uses **500 kbps CAN bus** over a shielded CAT5e/CAT6 RJ45 cable plugged into the `CAN` port on **Pack 1 (Master)** and the `BMS /`

`CAN` RJ45 port inside the Solis inverter.

RJ45 Pin	T568B Wire Color	Solis `BMS / CAN` Port	Easyway UNIV5200 `CAN` Port	Cable Wiring Rule
Pin 1 – 3	W-Orange / Orange / W-Green	NC / Enable GND / RS485-B	Reserved / Internal RS485	**Leave Disconnected***
Pin 4	**Solid Blue**	`CAN-H` **(CAN High)**	`CAN-H` **(CAN High)**	**Connect Pin 4 ↔ Pin 4**
Pin 5	**White / Blue**	`CAN-L` **(CAN Low)**	`CAN-L` **(CAN Low)**	**Connect Pin 5 ↔ Pin 5**
Pin 6 – 8	Green / W-Brown / Brown	RS485-A / **Pin 7: +12V Enable**	Reserved / Internal Bus	**Leave Disconnected***

⚠️ Field Pro-Tip: Isolate Pins 4 & 5 on Custom CAN Patch Cables!

On Solis hybrid/retrofit inverters, **Pin 7** of the BMS RJ45 socket carries `+12V Enable` and **Pins 3 & 6** carry RS-485 transceiver signals. While a standard T568B Ethernet cord aligns Pins 4 & 5, best electrician practice is to use the factory battery-to-inverter CAN cable or crimp a dedicated shielded CAT6 cable connecting **only Pin 4 (`CAN-H` Blue) and Pin 5 (`CAN-L` White/Blue)** (plus shield drain at the inverter end only) while leaving Pins 1–3 and 6–8 open so 12V never back-feeds into reserved BMS pins! *(Always verify the enclosed Easyway UNIV5200 hardware leaflet in case a regional sub-revision shifts CAN pins).*

2.3 Parallel Inter-Pack Link Wiring & Binary `ADD` DIP Switch Settings

When installing multiple `UNIV5200` packs in parallel, daisy-chain standard straight-through CAT5e/CAT6 patch cables from **Pack 1 (Master)** `RS485-B / Link Out` → **Pack 2 (Slave 1)** `RS485-A / Link In`, and from **Pack 2 Out → Pack 3 In** down the stack. Leave the `CAN` port empty on all Slave packs—**only Pack 1 (Master) connects to the Solis inverter!**

Before turning any battery ON, set the front-panel `ADD` **DIP switch block** (`ON = 1 / UP`, `OFF = 0 / DOWN`, read left-to-right from Switch 1 to Switch 4; if your module has a 6-pole switch block, keep Switches 5 & 6 `OFF` unless the enclosed leaflet specifies bus termination on the final pack):

UNIV5200 Pack Role	SW 1	SW 2	SW 3	SW 4	Binary Code	Notes
Single Pack / Master (Pack 1)	`OFF`	`OFF`	`OFF`	`OFF`	`0000`	Connects `CAN` to Solis
Slave 1 (Pack 2 — 10.44 kWh)	`ON`	`OFF`	`OFF`	`OFF`	`1000`	Link from Pack 1 Out
Slave 2 (Pack 3 — 15.66 kWh)	`OFF`	`ON`	`OFF`	`OFF`	`0100`	Link from Pack 2 Out
Slave 3 (Pack 4 — 20.88 kWh)	`ON`	`ON`	`OFF`	`OFF`	`1100`	Link from Pack 3 Out
Slave 4 (Pack 5 — 26.10 kWh)	`OFF`	`OFF`	`ON`	`OFF`	`0010`	Use Copper DC Busbar
Slave 5 (Pack 6 — 31.32 kWh)	`ON`	`OFF`	`ON`	`OFF`	`1010`	Use Copper DC Busbar
Slaves 6 – 15 (Packs 7 – 16)	Continue binary sequence: Pack 7 = `0110`, Pack 8 = `1110`, Pack 9 = `0001` … Pack 16 = `1111`				`0110..1111`	Up to 83.52 kWh total

2.4 Safe Pre-Charge Power-Up & Shutdown Sequence

> ⚡ **Mandatory Startup Order (Protects Inverter Capacitors & Prevents BMS Short Trips)**
>
> **Never** slam the DC MCCB closed onto an already-awake battery connected to cold inverter capacitors! Follow this exact sequence so the Easyway BMS soft-start pre-charge circuit energizes the Solis DC bus safely:
>
> 1. **Verify Polarity Cold:** With all batteries OFF, verify `BAT+` and `BAT-` continuity and zero reverse polarity using a multimeter, and double-check all `ADD` DIP switches.
> 2. **Close DC Path While BMS Is Asleep:** Switch individual `UNIV5200` pack power/breaker rockers to **ON** (if equipped) and switch the external **2-pole DC MCCB breaker ON** while the BMS LEDs are still off/standby.
> 3. **Wake Master BMS (Soft Pre-Charge):** Press and hold the `SW / POWER` **button** on **Pack 1 (Master)** for `2–3 seconds` until the LEDs sweep and the green `RUN` LED illuminates (parallel Slave packs wake automatically via the RS-485 link cable; confirm red `ALM` LED is OFF on every pack and measure `~51.2–53.5 Vdc` at the Solis DC lugs).
> 4. **Energize PV & AC Grid:** Turn **ON** the Solis DC PV isolator (on hybrid models), then turn **ON** the Solis AC Grid breaker.
>
> *Shutdown Order (Reverse):* Turn **AC Grid Breaker OFF** → Turn **PV DC Switch OFF** → Hold Master `SW` button 3s until all battery packs sleep → Switch **External DC MCCB OFF**.

Section 3: Connecting & Configuring an Eastron SDM72D-M or SDM630-M Smart Meter

Accurate whole-home bidirectional grid metering is essential so your Solis Hybrid or AC-Coupled Retrofit inverter and **GEMS** know every watt of rooftop solar export, household phase load, and EV charging demand. Eastron's 4-DIN-module 3-phase Modbus RTU meters—the **`SDM72D-M`** (and updated **`SDM72D-M-2`**) and **`SDM630-M`** (**`SDM630-Modbus V2 / V3`** direct-connected or **`SDM630MCT`** CT-connected)—share Eastron's IEEE-754 `Float32` register map (`0x0000–0x004A`) and integrate seamlessly with both Solis and GEMS.

3.1 Choosing Between Eastron SDM72D-M, SDM72D-M-2 & SDM630-M

Eastron Meter Model	Measurement Type	Supported Grid Systems	Best-Fit Application & Field Notes
Eastron SDM630-M *(`SDM630-Modbus V2 / V3`)*	**Direct-Connected** Up to `80 A / 100 A`	`3P4W` (`3×400V+N`) `3P3W` (`3×230V` Delta) `1P2W` (`1×230V`)	**Universal Reference** Officially mapped in Solis firmware (`Eastron 3P` / `SDM630`) and GEMS (`eastron_sdm630`). Ideal for standard 3-phase, single-phase, and Belgian `3×230V` Delta boards without Neutral!
Eastron SDM72D-M-2 *(and `SDM72D-M`)*	**Direct-Connected** Up to `80 A / 100 A`	`3P4W` (`3×400V+N`) *(`3P3W` on `-2` V2)*	**Cost-Effective 3P Direct** `SDM72D-M-2` (V2) exposes full per-phase `V / I / W` registers identical to `SDM630-M`. *(Note: Legacy V1 `SDM72D-M` requires Neutral `3P4W` and may omit per-phase V/I registers—prefer `SDM72D-M-2` or `SDM630-M` for Solis & 3-phase EV DLB).*
Eastron SDM630MCT *(Split-Core / Solid CT)*	**External CTs** `100A–600A / 5A` (or mV)	`3P4W` • `3P3W` • `1P2W`	**Best for Tight Retrofits & >100A** Clamps around existing incoming grid tails without disconnecting heavy utility feeders; requires setting the exact `CT Ratio` on the meter LCD.

> 💡 **Single-Phase Solis Inverter Installed on a 3-Phase Property**
> Even when installing a **1-phase** Solis inverter (`RAI-3K-48ES-5G`, `S6-EA1P-L`, `S5-EH1P-L`, or `S6-EH1P-L`) inside a home with a **3-phase utility supply**, always install a **3-phase meter (`SDM630-M` or `SDM72D-M-2`)** at the main grid entry! The 3-phase meter sums net import/export across `L1 + L2 + L3` so the 1-phase battery compensates loads on *all three* domestic phases under net phase metering.

3.2 Two Supported RS-485 Connection Topologies (Single-Master Bus Rule!)

> 🛑 **CRITICAL RS-485 RULE: Never Wire Two Modbus Masters to One Meter Simultaneously!**
> RS-485 Modbus RTU allows **strictly ONE Master** per bus! **Never** land both the Solis inverter `Meter` port cable AND a GEMS USB-RS485 cable onto the same `A+ / B-` terminals of a single `SDM72D-M` or `SDM630-M`—the two masters will collide and trigger `Meter_Comm_FAIL`. Choose **Topology 1** or **Topology 2** below:

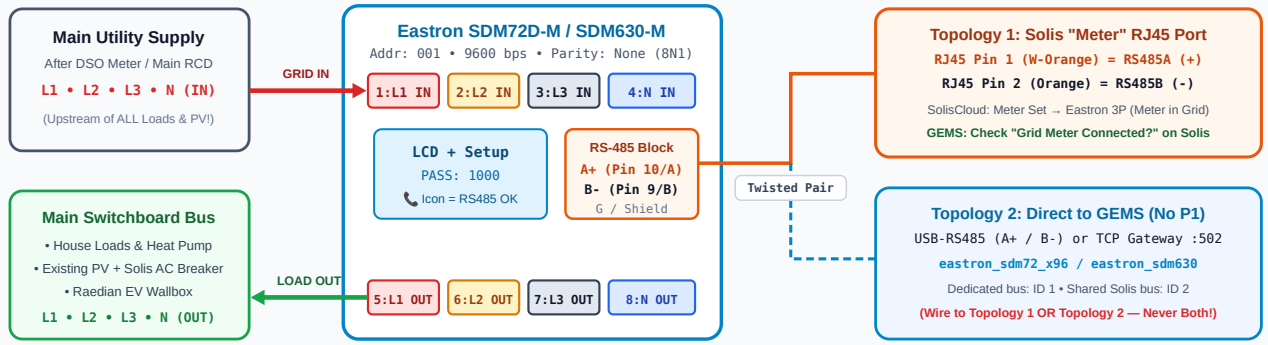
- **Topology 1 — Meter Wired to Solis `Meter` Port (Recommended Standard):** Wire the `SDM72D-M` / `SDM630-M` directly to the Solis inverter's **`Meter` RJ45 port**. Solis achieves sub-second hardware `Self-Use` tracking (even if your router or controller reboots) and automatically forwards live grid power (`33130–33131`) and 3-phase currents/voltages (`33251–33262`) to GEMS over the Solis Modbus link! *(If*

the home also has a digital utility meter, plug your USB P1 cable into GEMS alongside this setup for official DSO billing peaks).

- **Topology 2 — Meter Wired Directly to GEMS (Sites Without a Utility P1 Port or Dedicated Site Meter):** On analog Ferraris switchboards, unactivated P1 meters, or commercial sub-boards where GEMS needs direct `1–2s` per-phase metering for multi-inverter + **Raedian** EV Dynamic Load Balancing, wire a dedicated `SDM72D-M` or `SDM630-M` to GEMS via **USB-to-RS485** (or an RS-485 Ethernet Gateway on port `502`). *Multi-Drop Tip:* You can daisy-chain a GEMS-bound `SDM72D-M / SDM630-M` (set to **Modbus ID** `2`) on the same GEMS USB-RS485 cable as Solis Option A (**Modbus ID** `1`) at `9600 bps, 8N1` because GEMS is the sole Modbus master on that bus!

3.3 How to Connect Physically: AC Power, Placement & RS-485 Pinouts

Wiring Schematic: Eastron SDM72D-M / SDM630-M (100A Direct) ↔ Solis Meter Port (Topology 1) or GEMS (Topology 2)



1. **Switchboard Placement (Critical for Solis Retrofit & Hybrid):** Mount the `SDM72D-M` or `SDM630-M` on the DIN rail **immediately after the utility DSO meter / main RCD**—strictly **upstream** of all domestic circuit breakers, heat pumps, **Raedian** EV chargers, existing rooftop PV string inverters, and the Solis AC breaker. On AC-coupled Retrofit upgrades (`RAI-3K` / `S6-EA1P`), placing the meter upstream of the existing PV inverter guarantees all solar export passes through the meter and is captured into the Easyway `UNIV5200` stack.
2. **Main AC Power Wiring — Direct-Connected `SDM72D-M` / `SDM72D-M-2` & `SDM630-M` (Up to 100 A):**
   - **Top Cage Terminals (Grid Supply IN):** Wire incoming utility phases to `1 = L1 IN`, `2 = L2 IN`, `3 = L3 IN`, and `4 = N IN`.
   - **Bottom Cage Terminals (House / PV / Solis / EV Load OUT):** Feed your main switchboard distribution blocks from `5 = L1 OUT`, `6 = L2 OUT`, `7 = L3 OUT`, and `8 = N OUT`. Torque `10–25 mm²` copper ferrules cleanly to `2.5–3.0 Nm`.
   - **Belgian `3×230V` Delta (`3P3W`) & Single-Phase (`1P2W`) Notes:** On `3×230V` grids without Neutral, use `SDM630-M` or `SDM72D-M-2` and leave Neutral terminals `4 & 8` open. On 1-phase `1×230V` boards with `SDM630-M`, wire Line through `1 (L1 IN) → 5 (L1 OUT)` and Neutral through `4 (N IN) → 8 (N OUT)`. **Never swap Top (IN) and Bottom (OUT)** or import/export polarity will invert!
3. **External CT Wiring — `SDM630MCT` Variant Only:** Connect fused (`0.5A–2A`) voltage references to `V1/V2/V3/VN` (`Terminals 4, 3, 2, 1`) and wire `CT1 (Terminals 20=S1, 19=S2)`, `CT2 (18/17)`, and `CT3 (16/15)` matching `L1 / L2 / L3` phase-for-phase with CT arrows strictly aligned per the Solis/Eastron CT direction diagram.
4. **RS-485 Data Cable Pinout (`CAT5e / CAT6` or `LiYCY` Shielded Twisted Pair):**

Signal	Eastron Meter Terminal (`SDM72D-M` & `SDM630-M`)	Topology 1: Solis `Meter` Port (RJ45 T568B / 2-Pin Plug)	Topology 2: GEMS Controller (USB-RS485 / TCP Gateway)
`RS485-A (+)` *Data Non-Inverting*	`A` / `A+` / `485A` *(Pin `10` on SDM630-V2 Direct; Pin `14` on SDM630MCT; marked `A` on SDM72D-M)**	`RJ45 Pin 1` (White/Orange) *(or screw terminal `MTR485A`)*	Adapter `A+` / `D+` / `T/R+`
`RS485-B (-)` *Data Inverting*	`B` / `B-` / `485B` *(Pin `9` on SDM630-V2 Direct; Pin `13` on SDM630MCT; marked `B` on SDM72D-M)**	`RJ45 Pin 2` (Solid Orange) *(or screw terminal `MTR485B`)*	Adapter `B-` / `D-` / `T/R-`
Shield / Ground *& Other RJ45 Pins*	`G` / `GND` *(Pin `11/12` or marked `G`; **ignore `+ / -` Pulse outputs!**)*	`Leave RJ45 Pins 3 – 8 Open` *(Ground shield at one end only)*	Adapter `GND` *(`120 Ω` across A–B if > 30 m)*

*Always verify the laser-printed terminal legend under your meter's hinged cover so RS-485 `A / B` are never confused with SO Pulse Output `+ / -` terminals.

3.4 How to Configure: Meter LCD Menu, SolisCloud App & GEMS

Step 1: Configure Front-Panel Communication Settings on the SDM72D-M / SDM630-M (Password: 1000)

- **Unlocking `SDM630-M` / `SDM630MCT`:** Press and hold the `E` **(Enter)** button for `3 seconds` until `PASS` (`0000`) appears. Use the scroll buttons (`U / P`) to enter `1000` (press `E` to advance each digit, then hold `E` for 3s to open the Setup menu).
- **Unlocking `SDM72D-M` / `SDM72D-M-2`:** Press and **long-hold** the front setup button (`~3 seconds`) until the password prompt appears. Use **short presses** to increment digits and **long presses** to advance digits to enter `1000`, then long-press to enter Setup.

Meter LCD Menu Item	Required Value	Configuration Rule & Verification
Modbus Address (`Addr` / `Add`)	`001` *(Default)* (`002` *on shared GEMS bus)*	**Must be `001`** when wired to the Solis `Meter` port or a dedicated GEMS adapter. Set to `002` *only* when daisy-chained on the same GEMS USB-RS485 cable as a Solis inverter using ID `1`.
Baud Rate (`bAud` / `baud`)	`9600` (`9.6k`)	Matches Solis hardware `Meter` port speed and GEMS default serial discovery baud rate (`9600 bps`).
Parity & Framing (`PrtY` / `Par`)	`nO / none` (`8N1`)	**Always double-check!** Some utility wholesale batches ship set to `Even` parity—change to `nO` / `none` (1 stop bit) so Solis and GEMS communicate cleanly.
Grid Wiring System (`SYS` — `SDM630` / `72D-M-2`)	`3P4` (`3×400V+N`) `3P3` (`3×230V` Delta) `1P2` (`1×230V`)	Select `3P4` on standard 3-phase+N boards, `3P3` on Belgian `3×230V` Delta switchboards without Neutral, or `1P2` on single-phase supplies.
CT Primary Ratio (`Ct` — `SDM630MCT` only)	Match CTs (e.g. `20` or `40`)	For `100A/5A` CTs set `20`; for `200A/5A` CTs set `40`. *(Skip on direct-connected `SDM72D-M` & `SDM630-M`).*

Hold `E` (or wait 10 seconds idle on `SDM72D-M`) to exit Setup back to live `kW` measurements. Once RS-485 polling starts, verify the small **telephone handset (📞) icon** appears on the Eastron LCD!

Step 2: Configure the Solis Inverter in SolisCloud App (When Using Topology 1 — Solis Meter Port)

1. Connect locally via **SolisCloud App → Me → More Tools → Local Operation → Connect with Bluetooth** (Installer login).
2. Open `Advanced Settings → Storage Energy Set → Meter Set (Meter Select)`:
   - **Meter Type / Model:** Select `Eastron 3P` (or `SDM630` on LCD menus). Because `SDM72D-M` / `SDM72D-M-2` and `SDM630-M` share identical IEEE-754 Float32 active power and phase registers, selecting `Eastron 3P` / `SDM630` works for **both** models! *(Only select `Eastron 1P` if using a single-phase SDM120/SDM230).*
   - **Meter Placement:** Select `Meter in Grid` (confirms meter sits at the utility boundary).
3. Open **Information → Meter Info** in SolisCloud: confirm zero `Meter_Comm_FAIL` alarms and check that switching on a ~2 kW household kettle registers ~2 kW grid import in the expected direction.

Step 3: Configure the Meter in GEMS (`Settings → Devices`)

- **If Wired to Solis `Meter` Port (Topology 1):** Do *not* add a separate Eastron device card. Simply enable `Grid Meter Connected? = Checked (✓)` inside your Solis Hybrid or Solis Retrofit device configuration in GEMS—GEMS automatically reads live grid watts and 3-phase Eastron currents directly through the Solis inverter!
- **If Wired Directly to GEMS (Topology 2 — No P1 Utility Meter):** Unlock **Certified Installer Mode** (🔒), go to **Settings → Devices → + Add Device → Smart Meter** (or click **Auto-Discover Serial / RS-485 Ports** to prefill automatically!), and configure:
  - **Template:** Choose `Eastron SDM72D / SMART X96 3-Phase Meter` (`eastron_sdm72_x96`) for `SDM72D-M` / `SDM72D-M-2`, or `Eastron SDM630 3-Phase Meter` (`eastron_sdm630`) for `SDM630-M` / `SDM630MCT`.
  - **Endpoint & Serial Settings:** Select the green persistent `/dev/serial/by-id/...` pill (`Baud = 9600`, `Parity = None`) or enter your Ethernet RS-485 gateway IP on `Port 502`.
  - **Modbus ID & Poll Interval:** Set `Modbus ID = 1` (or `2` if sharing the same USB-RS485 bus with Solis at ID `1`), `Poll Interval = 2s`, and `Phase Wiring = L1-L2-L3`, then click **Save Device** and verify `• Online` with live `L1 / L2 / L3` currents and voltages under **Debug**!

Section 4: Connecting Solis to GEMS (RS-485 & LAN Stick Topologies)

To achieve fast, zero-cloud closed-loop battery dispatch and solar curtailment, connect your Solis inverter to GEMS over local Modbus RTU (RS-485) or local Modbus TCP. Choose one of the two verified topologies below:

GEMS + Solis + Easyway UNIV5200 Communication Topologies



Connection Option	Physical Wiring & Pinout Details	Field Verdict
Option A (Best): Direct USB-to-RS485 or Industrial Ethernet Gateway	• **Solis S6 Dedicated Internal** `RS485 / Modbus` **RJ45 / Terminal Port:** Connect shielded twisted pair to the dedicated internal EMS/Cascade RS-485 port so your external SolisCloud logger stick stays plugged into the 4-pin COM port uninterrupted! • **Solis 4-Pin Circular** `COM` **Port** (on S5 / RAI or via Y-splitter): **Pin 3 (Yellow) =** `RS485A (+)`, **Pin 4 (Blue) =** `RS485B (-)`, optional **Pin 2 (Black) =** `GND`. **NEVER connect Pin 1 (+5V Red)** to a USB-RS485 adapter!	**Gold Standard** Zero Wi-Fi jitter, deterministic `2–3s` closed-loop control, and works independently of SolisCloud!
Option B: Solis `S2-WL-ST` **(Wi-Fi + LAN Stick) or** `DLS-L`	Plug an Ethernet RJ45 cable into the bottom of the `S2-WL-ST` dual Wi-Fi/LAN stick, reserve a static DHCP IP on your router, and connect GEMS via Modbus TCP on port `502` (Slave ID `1`).	**Clean No-Cable LAN** Supports simultaneous SolisCloud + single local Modbus TCP client on port `502` (poll every `5s`).
Legacy Cloud-Only `S3-WiFi-ST` **Stick**	Entry-level Wi-Fi-only sticks manufactured strictly for outbound 5-minute SolisCloud pushes often disable raw Modbus TCP port `502` in recent firmware.	**Not Recommended** Upgrade the stick to a wired `S2-WL-ST` or wire Option A (USB-RS485 / Ethernet RS-485 gateway).

Section 5: Exact Settings in the SolisCloud App & Inverter HMI

Standard homeowner plant views in SolisCloud only display 5-minute charts. To configure battery BMS handshakes, Eastron smart meter selection, storage modes, and grid-charging permissions, connect locally using **SolisCloud Local Debugging (Bluetooth)** or the inverter's front LCD panel.

5.1 Entering Local Commissioning in the SolisCloud App

1. Stand within 3 meters of the Solis inverter with **Bluetooth enabled** on your smartphone.
2. Open the **SolisCloud App**. On the login screen (or bottom tab bar under **Me → More Tools**), tap **Local Operation / Local Debugging → Connect with Bluetooth**.
3. Select your inverter by its serial number, choose **Installer**, and authenticate using your authorized Solis Installer credentials (or on LCD-equipped S5/RAI models, go to **Advanced Settings** and enter the Solis Installer passcode from your commissioning card).

5.2 Complete Solis Parameter Checklist (SolisCloud Local Commissioning)

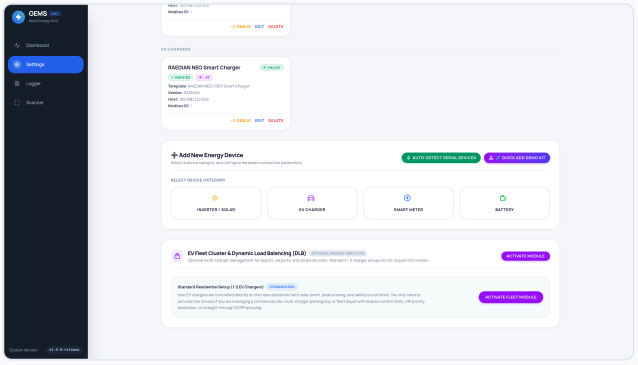
SolisCloud Menu & Parameter	Required Setting	Why This Exact Setting Is Critical for UNIV5200 + GEMS
`Storage Energy Set → Battery Select` **Battery Type & Brand Profile**	**Lithium Battery** → **PYLON LV** *(or `Lithium Battery LV` / `AoBo`)*	Establishes closed-loop 500 kbps CAN communication with the Easyway `UNIV5200` BMS over Pins 4 & 5. **Never** select *Lead-Acid* or *No Battery*, and avoid profiles marked *(RS485)*. Verify immediately under **Information → BMS Info** that status reads **Normal** with live SOC % and current limits!
`Battery Select` **Overdischarge SOC**	**10 %** *(Standard)* **15 % – 20 %** *(With Backup Load)*	Matches Easyway UNIV5200's rated **90% Depth of Discharge**. Stops household discharge smoothly at 10% SOC (or reserves 15–20% if critical loads are wired to the Solis `AC-Backup` port).
`Battery Select` **Forcecharge SOC**	**5 %** **Keep ≥5% below Overdischarge!**	**Crucial Solis anti-bounce rule!** If `Forcecharge SOC` is set too close to `Overdischarge SOC` (e.g. 9% vs 10%), natural 1% overnight BMS standby drift triggers loud, unwanted emergency AC grid recharges every night! Setting `10% / 5%` eliminates bounce-charging completely.
`Battery Select` **Max Charge / Discharge Current**	**50 A × Pack Count** *(Capped at Inverter Max A)*	Set `50 A` for 1× UNIV5200 (~2.56 kW), `100 A` for 2× UNIV5200 (~5.12 kW), or inverter max (e.g. `125–190 A`) for 3+ packs. GEMS also reads live BMS dynamic current limits every cycle.
`Storage Mode Select` **Work Mode**	**Self-Use** — ON **Time of Charging (TOU) — OFF** **Feed-in Priority — OFF**	Sets autonomous solar self-consumption as the hardware baseline! Leave all 6 Solis internal *Time-of-Use* clock slots disabled (`00:00–00:00`) so internal timers never fight GEMS's real-time remote battery control.
`Storage Mode Select` **Allow Grid Charging** *(Battery Charge from Grid)*	**Allow / Enable**	**Mandatory for dynamic spot tariffs & Winter Fallback!** When *Self-Use* is active and TOU timers are OFF, enabling *Allow Grid Charging* will **not** charge from AC on its own—it simply grants hardware permission when GEMS commands cheap/negative grid charging.
`Meter Set (Meter Select)` **Meter Type & Placement**	**Eastron 3P** (`SDM630` / `SDM72D-M`) *(or `Acrel 1P / 3P`)* Placement: **Meter in Grid**	Selects the IEEE-754 Float32 protocol for both **Eastron** **`SDM630-M`** and **`SDM72D-M`** / **`SDM72D-M-2`** at `Addr 001, 9600 8N1` (see Section 3) for autonomous `Self-Use` fallback and live 3-phase current sensing.
`Export Power Set (Internal EPM)` **Backflow Power / Export Limit**	**OFF (Unlimited)** *(Unless legally mandated by DSO)*	Keeping static EPM *OFF* allows profitable solar export and Flemish Green Certificates, while letting GEMS dynamically curtail hybrid PV down to 0 W only during negative-price hours!
`Communication / Modbus` **Slave Address & Baud Rate**	Address: **1** Baud: **9600 bps, 8N1**	Matches GEMS default discovery and polling parameters. (Use `1, 2, 3...` if daisy-chaining multiple Solis inverters or a GEMS-bound Eastron meter at ID `2` on one RS-485 bus).

Section 6: Onboarding Solis Hybrid vs. Solis Retrofit in GEMS

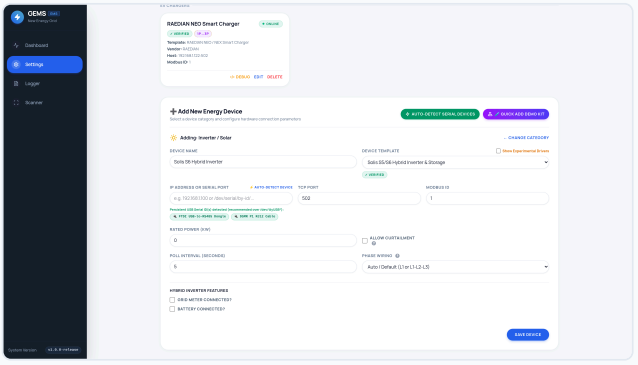
Open the GEMS web interface (`https://gems.local` or your controller's LAN IP), navigate to **Settings**, click the **padlock icon** 🔒 in the top-right header, and enter your **Installer Access PIN** to unlock **Certified Installer Mode**.

6.1 Picking the Right GEMS Template: Hybrid vs. AC-Coupled Retrofit

- **Case 1 — Solis Hybrid Inverter with PV Strings + UNIV5200 (`S5-EH1P`, `S6-EH1P`, `S6-EH3P-L`):** Onboard your primary site meter first if present (`p1_serial`, or direct `eastron_sdm72_x96` / `eastron_sdm630` per Section 3 Topology 2), then onboard the inverter under **Inverter / Solar** → **`Solis S5/S6 Hybrid Inverter`** (`solis_hybrid`). GEMS reads pure DC solar PV input power separately from AC inverter output so nighttime Easyway battery discharge never appears as false solar generation, and unlocks `0.1%` stepless PV curtailment!
- **Case 2 — Solis AC-Coupled Retrofit Inverter + UNIV5200 (`RAI-3K-48ES-5G`, `S6-EA1P-L`) Alongside Existing PV:**
  1. Onboard your utility **P1 Serial Meter** (`p1_serial`) — or a dedicated GEMS-wired **Eastron** **`SDM72D-M`** / **`SDM630-M`** (`eastron_sdm72_x96` / `eastron_sdm630`) if no P1 port is available — under **Smart Meter**.
  2. Onboard your **existing rooftop solar string inverter** (e.g. Solis string, SMA, Huawei, Enerlution, etc.) under **Inverter / Solar** with `Battery Connected? = Unchecked`.
  3. Onboard the AC-coupled Solis unit under **Battery Storage** → **`Solis Retrofit Storage System`** (`solis_retrofit_battery`) — or under **Inverter / Solar** → **`Solis Retrofit Inverter (AC Storage)`** (`solis_retrofit`). GEMS strictly sets PV string generation on the retrofit unit to `0 W` while giving you full bidirectional Easyway `UNIV5200` peak shaving, solar surplus charging, and spot arbitrage!



1. Choose Inverter / Solar (Hybrid) or Battery Storage (Retrofit)



2. Enter Endpoint, Rated kW, Phase Wiring & Total UNIV5200 kWh

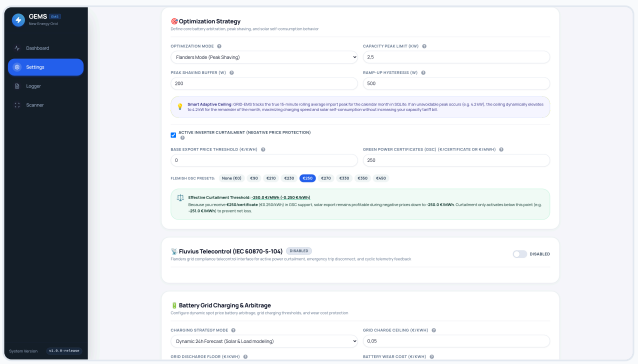
6.2 Field-by-Field GEMS Device Onboarding Reference

GEMS Device Field	Recommended Value	Technical Purpose & Best Practice
Template	`Solis S5/S6 Hybrid Inverter` *or* `Solis Retrofit Storage System`	Selects verified primary reference driver with 80 ms bus turnaround pacing, live BMS limit reader, and 60s volatile RC keepalive.
IP Address or Serial Port	`/dev/serial/by-id/...` *(Option A)* or Static IP of `S2-WL-ST` *(Option B)*	Always click the green persistent `/dev/serial/by-id/...` pill when using USB-RS485 (`9600 bps` , `None` parity) so ports never swap across reboots.
TCP Port & Modbus ID	Port: `502` • Modbus ID: `1`	Standard Modbus TCP port for `S2-WL-ST` and RS-485 Ethernet gateways; matches Solis Slave Address `1` .
Rated Power (kW)	Exact Inverter AC Rating *(e.g.* `3.0` , `3.6` , `5.0` , `6.0` , `10.0` *)*	Used on Hybrid models to scale `0.1%` active solar curtailment accurately and cap maximum dispatchable AC power.
Allow Curtailment	**Checked (✓)** on Hybrid PV **Unchecked (☐)** on AC Retrofit	Enable on Solis Hybrid inverters (and on any separate solar inverter in Retrofit homes) to throttle PV during negative export hours.
Poll Interval (seconds)	**2 – 3 s** *(Direct RS-485 / Gateway)* **5 s** *(S2-WL-ST LAN Stick)*	Direct RS-485 supports rapid 2–3s sub-cycle responsiveness; `5s` keeps `S2-WL-ST` sticks completely congestion-free alongside SolisCloud.
Phase Wiring	Exact Phase: `L1` , `L2` , or `L3` (1P) or `L1-L2-L3` (3P `S6-EH3P-L`)	**Vital for 1-phase RAI / S6-EA1P / S6-EH1P on 3-phase homes!** Identifies which exact domestic phase receives battery peak-shaving & fuse relief.
Grid Meter Connected?	**Checked (✓)** if SDM72D-M / SDM630-M / Acrel wired to Solis **Unchecked (☐)** if no meter on Solis	Check ✓ whenever an Eastron `SDM72D-M` , `SDM630-M` , or Acrel meter is wired to the Solis `Meter` port (Section 3 Topology 1). Leave ☐ if metering is wired solely to GEMS.
Battery Connected? & Capacity (kWh)	**Checked (✓)** + Exact Total kWh: 1× = `5.2` • 2× = `10.4` • 3× = `15.7` • 4× = `20.9`	Drives GEMS's 24-hour dynamic arbitrage solver, SOC headroom balancing, and Winter Fallback charge duration calculations.

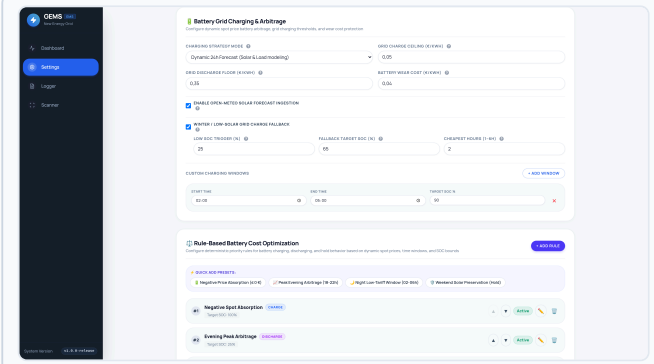
Section 7: Optimizing GEMS Strategy for Solis & Easyway UNIV5200

7.1 Location, Solar Forecast & Grid Safety Limits (`Settings → Strategy`)

1. **Enable Open-Meteo Solar Forecast:** Enter your installation street address under **Geographic Location**, click **Lookup Lat/Lon**, and ensure **Enable Open-Meteo Solar Forecast Ingestion** is `Checked (✓)` in the Battery card. GEMS will automatically hold off overnight grid charging when tomorrow's solar forecast is strong enough to fill your Easyway `UNIV5200` stack for free!
2. **Configure Main Breaker & Grid Topology:** Set your exact main fuse rating (e.g. `25 A`, `32 A`, or `40 A`) and switchboard topology (`Single Phase 230V`, `Three Phase 400V`, or Belgian `Three Phase 230V Delta`). Keep **Meter Offline Fallback Current** at `0 A — Pause EV Charging`.



Regional Optimization Mode, Flanders Peak Shaving & GSC Shield



Dynamic 24h Forecast, 0.04 €/kWh Wear Protection & Winter Fallback

7.2 Regional Optimization Recipes

🇧🇪 Recipe A: Flanders (Peak Shaving)

- **Mode:** `Flanders Mode (Peak Shaving)`
- **Capacity Peak Limit:** `2.5 kW` (standard) or `3.0–3.5 kW` (heat pump + EV). *Smart Adaptive Ceiling* auto-raises if a higher monthly peak occurs.
- **Peak Shaving Buffer:** `150 – 200 W`
- **Active Curtailment:** `Checked (✓)`, Base `0.00 €/kWh`
- **Flemish GSC Shield:** Set `€0` (modern PV) or exact certificate rate (e.g. `€250–€450/MWh`)—especially common on Solis Retrofit upgrades of legacy certificate-earning solar roofs!

🇳🇱 Recipe B: Netherlands (Saldering)

- **Mode:** `Netherlands Mode (Smart Saldering)`
- **Active Curtailment:** `Checked (✓)`
- **Base Export Price Threshold:** `0.00 €/kWh` (or your supplier's feed-in levy threshold).
- **How It Works:** Routes daytime PV surplus into Easyway `UNIV5200` packs and EV charging first to avoid grid feed-in levies, then curtails PV smoothly once storage reaches 100% SOC during negative-return hours.

🌿 Recipe C: Fixed / Dual-Tariff Eco

- **Mode:** `Eco (Max Self-Consumption)`
- **Active Curtailment:** `Unchecked (□)` (unless on a contract with negative export charges).
- **Charging Strategy:** `Price Threshold Only` (`-0.01 €/kWh`) or `Super-Dal Only` for cheap off-peak night tariffs.
- **How It Works:** Maximizes solar self-sufficiency and 8,000+ cycle $LiFePO_4$ lifespan with zero unnecessary grid cycling.

7.3 Optimal UNIV5200 Arbitrage, Longevity & EV Coordination Settings

GEMS Strategy Parameter	Optimal Value for UNIV5200	Operational Benefit
Charging Strategy Mode	**Dynamic 24h Forecast**	Optimizes charge/discharge across 24h Day-Ahead tariffs, solar forecast, and 30-day Weekday/Weekend load profiles while respecting live UNIV5200 BMS limits.
Battery Wear Cost (€/kWh)	**0.04 €/kWh**	Accounts for ~12–15% 48V AC-DC-AC round-trip conversion losses + $LiFePO_4$ cycle wear so GEMS only executes grid arbitrage when net profit is genuinely positive.
Grid Charge Ceiling / Discharge Floor	Ceiling: `0.08 €/kWh` Floor: `0.35 €/kWh`	Absorbs ultra-cheap or negative daytime/overnight grid power while reserving stored UNIV5200 energy for household peak consumption.
Winter / Low-Solar Grid Charge Fallback	**Checked (✓)** `25%` → `65%` (`2–3h`)	Automatically tops up UNIV5200 packs from `≤25%` to `65% SOC` during the 2–3 cheapest night hours in dark winter weather without exceeding your Flanders 15-min peak target!
Overnight EV Charging Protection *(e.g. with **Raedian** Wallbox)*	Add Custom Battery Rule: **Hold (Idle / No Export)**	Prevents a night EV boost session on your **Raedian NEO / NEX / Gemini** wallbox from draining the Easyway `UNIV5200` battery—keeping 100% battery reserve for the morning household peak!

Section 8: Hardware Safeguards, Handover Checklist & Troubleshooting

8.1 How GEMS Protects Solis Flash Memory & Easyway UNIV5200 Cells

- **Zero-EEPROM-Wear Volatile Remote Control:** Naive controllers repeatedly rewrite Solis's non-volatile storage mode flash register every few seconds, wearing out inverter EEPROM within a year. GEMS checks baseline storage permissions *once* and executes all real-time charge, discharge, and hold modulation exclusively via Solis's **volatile RAM Remote Control registers** with a 60-second watchdog keepalive—causing zero flash memory wear!
- **Live BMS Dynamic Current Clamping:** Cold winter utility rooms (< 10 °C) and high State-of-Charge (> 90% SOC) cause the Easyway `UNIV5200` BMS to lower permissible charge current automatically. GEMS reads live BMS charge/discharge amp limits and pack voltage directly from the Solis inverter and clamps setpoints smoothly.
- **Periodic 100% $LiFePO_4$ Top-Balancing:** Because $LiFePO_4$ cells have a very flat voltage curve between 20% and 90% SOC, allowing the UNIV5200 stack to reach `100% SOC` from solar (or occasional cheap grid hours) at least once every 1–2 weeks keeps the BMS SOC coulomb counter razor-sharp across all parallel packs.

8.2 6-Point Post-Commissioning Handover Checklist

✅ **Mandatory Site Verification Before Handover**

1. **BMS & Smart Meter Handshake:** Confirm zero red `ALM` LEDs on all Easyway `UNIV5200` packs, Solis `BMS Info = Normal`, and active telephone handset (`📞`) icon on the Eastron `SDM72D-M` / `SDM630-M` LCD with zero `Meter_Comm_FAIL` alarms.
2. **GEMS Device Online & Debug Check:** In **Settings** → **Connected Devices**, verify Solis (and any GEMS-wired Eastron meter) shows `• Online`. Click **Debug** and confirm accurate Voltage (~230 V), SOC %, per-phase currents, and Battery/Grid watts.
3. **Kettle Load Polarity Test:** Switch on a ~2 kW household appliance. Confirm Home Consumption rises by ~2 kW on the GEMS Dashboard and that Solis seamlessly ramps Easyway discharge to cover the load in `Eco / Self-Use` mode.
4. **Logger Cleanliness Check:** Open the **Logger** tab for 2 minutes and verify clean polling with zero CRC errors, timeouts, or socket resets.
5. **Export JSON Backup:** In **Settings** → **System Info**, click **Export Configuration Backup (.json)** for site documentation.
6. **Lock Certified Installer Mode:** Click the padlock icon 🔒 in the Settings header so critical battery capacities, phase wiring, and fuse limits are protected.

8.3 Field Troubleshooting Matrix

Observed Symptom	Probable Root Cause	Field Corrective Action
Solis alarm `CAN_Comm_FAIL` or `BatName-FAIL`	Wrong Battery Select profile, CAN cable plugged into Slave or RS485 port, or wrong RJ45 pinout.	Plug CAN cable strictly between **Pack 1 (Master, `ADD=0000`)** **`CAN`** **port** and Solis **`BMS`** **port** (Pin 4 `CAN-H`, Pin 5 `CAN-L` isolated). Select `Lithium Battery → PYLON LV` in SolisCloud.
Solis alarm `Meter_Comm_FAIL` (or Eastron offline in GEMS)	RS-485 `A/B` swapped, meter set to `Even` parity or wrong Slave ID, wires landed on Pulse `+/-`, or two masters wired to one meter.	Unlock meter (`PASS 1000`): set `Addr = 001`, `bAud = 9600`, `PrtY = nO / none` (`8N1`). Verify Solis RJ45 **Pin 1 (W-Orange) → Meter `A+`** and **Pin 2 (Orange) → Meter `B-`**, select `Eastron 3P` in SolisCloud, and never attach two masters to one meter.
Only 1 pack recognized in a 2–4 pack UNIV5200 stack	Duplicate `ADD` DIP switches, DIP changed while battery was ON, or inter-pack RS485 link swapped.	Power all packs completely OFF, set unique binary `ADD` codes (`0000`, `1000`, `0100`...), verify `RS485-B Out → RS485-A In` daisy-chain, and cold-boot Pack 1 Master.
Battery charges briefly from grid every night at 10% SOC	Solis `Forcecharge SOC` set equal to or within 1–2% of `Overdischarge SOC`.	In SolisCloud `Battery Select`, set `Overdischarge SOC = 10%` and lower `Forcecharge SOC = 5%` to leave a clean 5% hysteresis deadband.
5 kW Solis inverter caps battery charge/discharge at ~2.55 kW	Only a single `UNIV5200` pack is installed (single-pack BMS recommends `50 A × 51.2 V ≈ 2.56 kW`).	Normal single-pack BMS protection! Add a second parallel `UNIV5200` pack (`100 A ≈ 5.12 kW`) if full 5 kW continuous charge/discharge is desired.
GEMS reads SOC %, but forced grid charging does not start	`Allow Grid Charging` disabled in SolisCloud, or conflicting Solis internal TOU schedule active.	In SolisCloud `Storage Mode Select`, enable `Self-Use`, set `Allow Grid Charging = Allow`, and turn all internal *Time of Charging* slots `OFF`.
`i/o timeout` or `Connection refused` on port `502` with Wi-Fi stick	Site uses an outbound cloud-only `S3-WiFi-ST` stick (drops all inbound port `502` packets even while answering LAN ARP), `S2-WL-ST` firmware predates local Modbus TCP (`< 100121cf` / `100141dx`), or second client holds port `502`.	Connect GEMS via direct USB-RS485 / RS-485 Ethernet gateway (Option A), replace `S3-WiFi-ST` with wired Ethernet **`S2-WL-ST`**, or request Solis Support to push the Modbus TCP datalogger firmware update onto an existing `S2-WL-ST`.
Battery discharges to grid (or charges from grid) when house load turns ON	`SDM72D-M / SDM630-M` Top (`1–4` IN) & Bottom (`5–8` OUT) swapped, or `SDM630MCT` / Acrel CT clamp reversed.	Ensure utility supply enters **Top `1–4` (IN)** and house/PV/Solis exit **Bottom `5–8` (OUT)**; on CT meters, verify CT arrow orientation and `L1/L2/L3` phase matching.
Nighttime battery discharge shows up as Solar PV on Retrofit	AC-coupled `RAI-3K` or `S6-EA1P` was onboarded using `solis_hybrid` instead of `solis_retrofit`.	Edit the device in **Settings → Devices** and switch Template to `Solis Retrofit Storage System` (or `Solis Retrofit Inverter`).

> 📞 **Fleet Monitoring & Installer Support**
> Pair your GEMS unit with **GRID-EMS-SERVER** (`https://ems.newenergygrid.com`) under **Settings → Cloud Server** for remote 24-hour forecast diagnostics, live BMS telemetry verification, and centralized multi-site management by **New Energy Grid** & **The Charge Grid**.